

SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE:CSA1513

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**

Question Count: 4

Code of Conduct

I B Sandhya(192311292)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sandhya

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

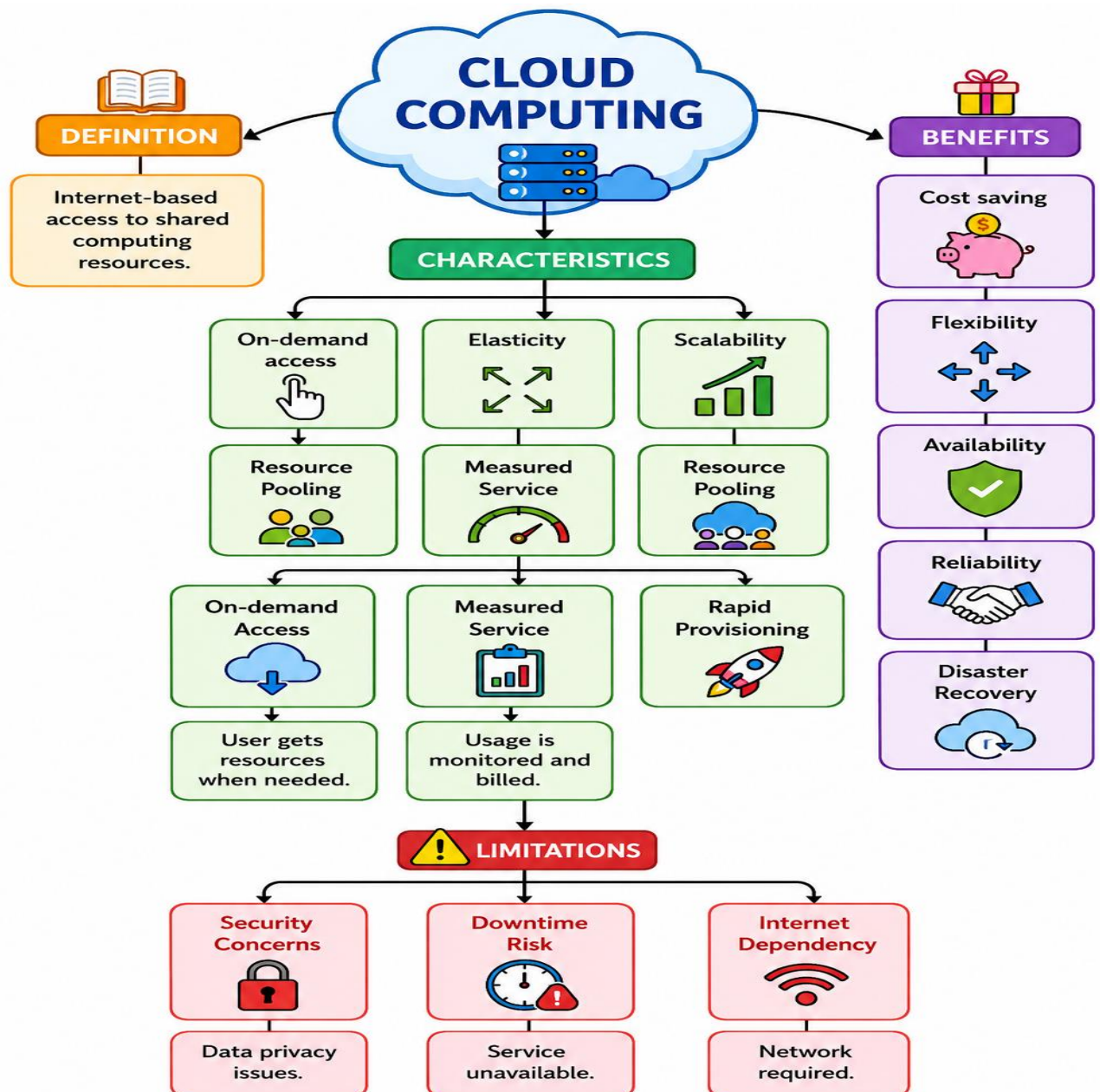
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

ANSWERS

1. Concept Map 1: Cloud Computing Fundamentals

Main Concept: ☁ Cloud Computing

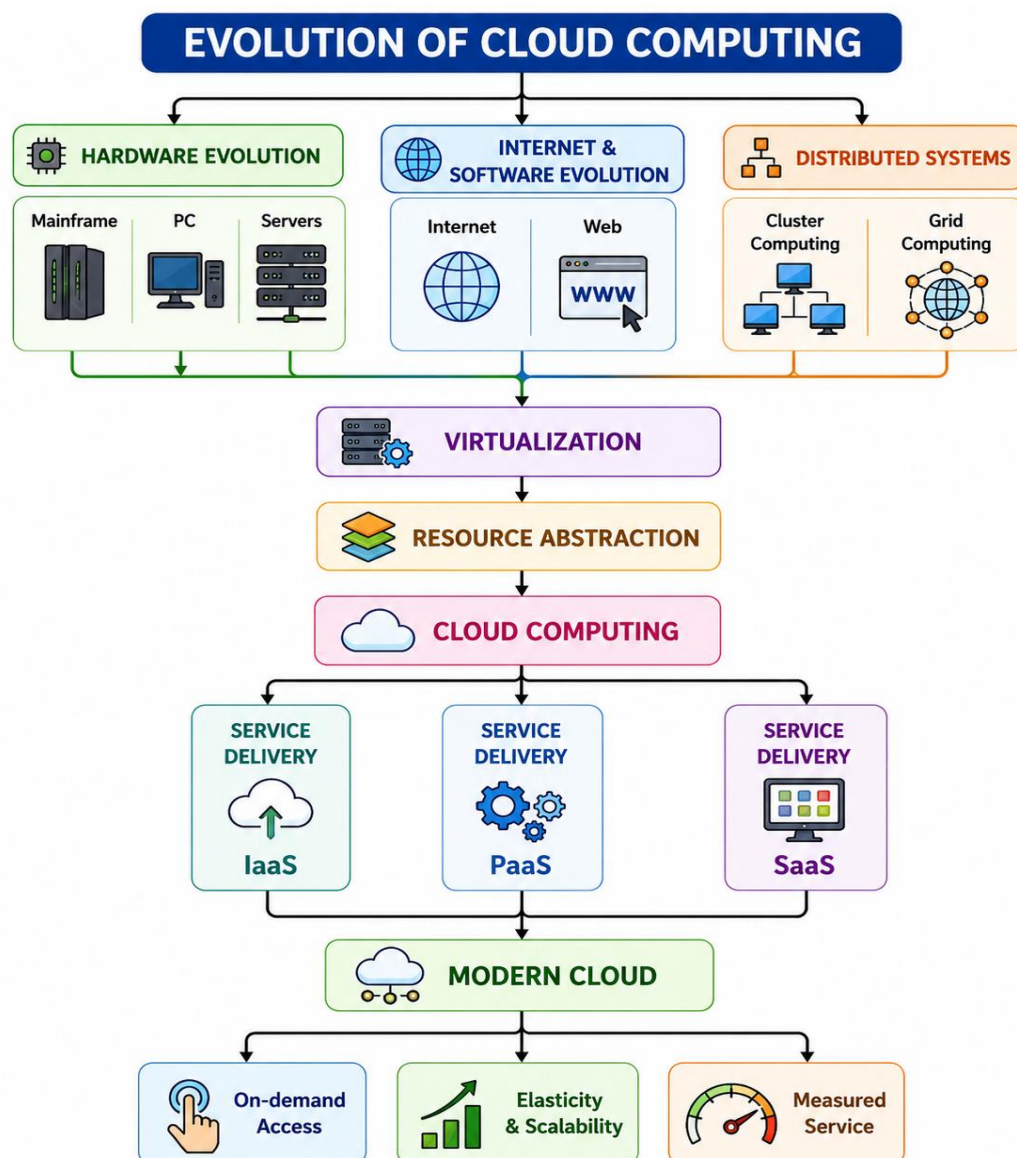


Key Linking Labels:

- Cloud Computing **provides** → On-demand access
- Cloud Computing **uses** → Resource pooling
- Cloud Computing **supports** → Elasticity and scalability
- Cloud Computing **offers** → Cost savings and flexibility
- Cloud Computing **has limitations such as** → Security, downtime, internet dependency

2. Concept Map 2: Evolution of Cloud Computing

Main Concept: 🌐 Evolution of Cloud Computing



Evolution Flow:

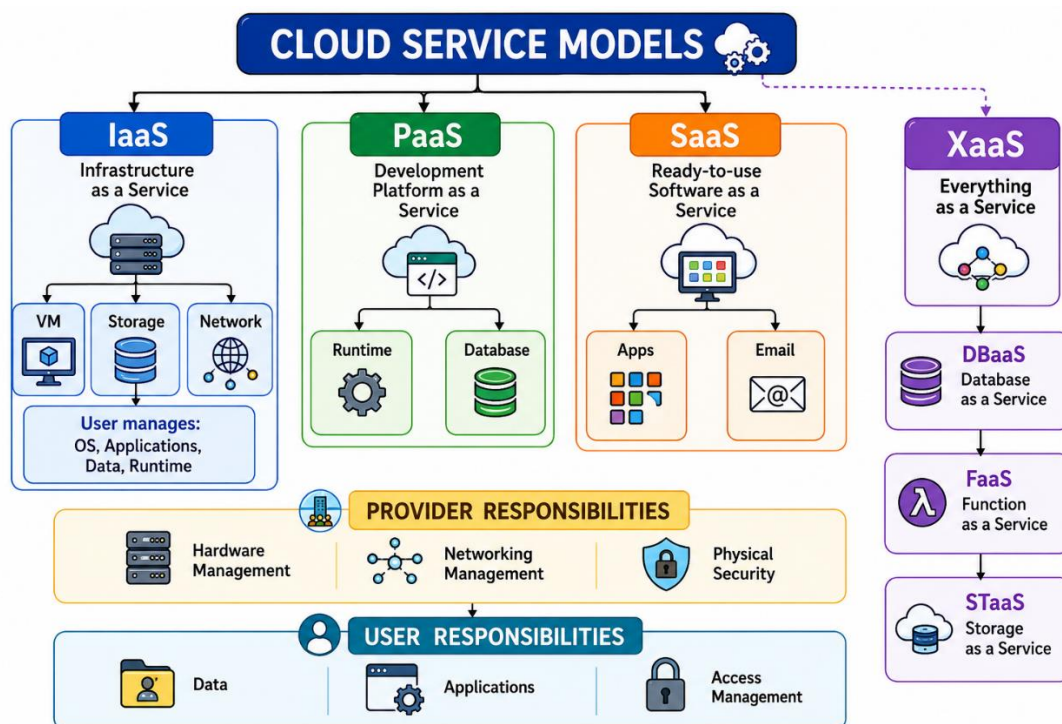
Mainframes → Personal Computers → Client-Server → Internet → Distributed Systems → Virtualization → Cloud Computing → IaaS/PaaS/SaaS

Key Linking Labels:

- Hardware evolution **increased** → Computing power
- Internet evolution **enabled** → Remote access
- Distributed systems **connected** → Multiple resources
- Virtualization **enabled** → Resource sharing
- Cloud computing **delivers** → Computing as a service

3. Concept Map 3: Cloud Service Models

Main Concept: ☁ Cloud Service Models



Responsibility Hierarchy

Service Model	Provider Manages	User Manages
IaaS	Hardware, storage, networking, virtualization	OS, applications, data
PaaS	Infrastructure, OS, runtime, platform	Applications, data
SaaS	Infrastructure, platform, application	Data and user settings
XaaS	Service infrastructure	Service usage and data

Logical Hierarchy:

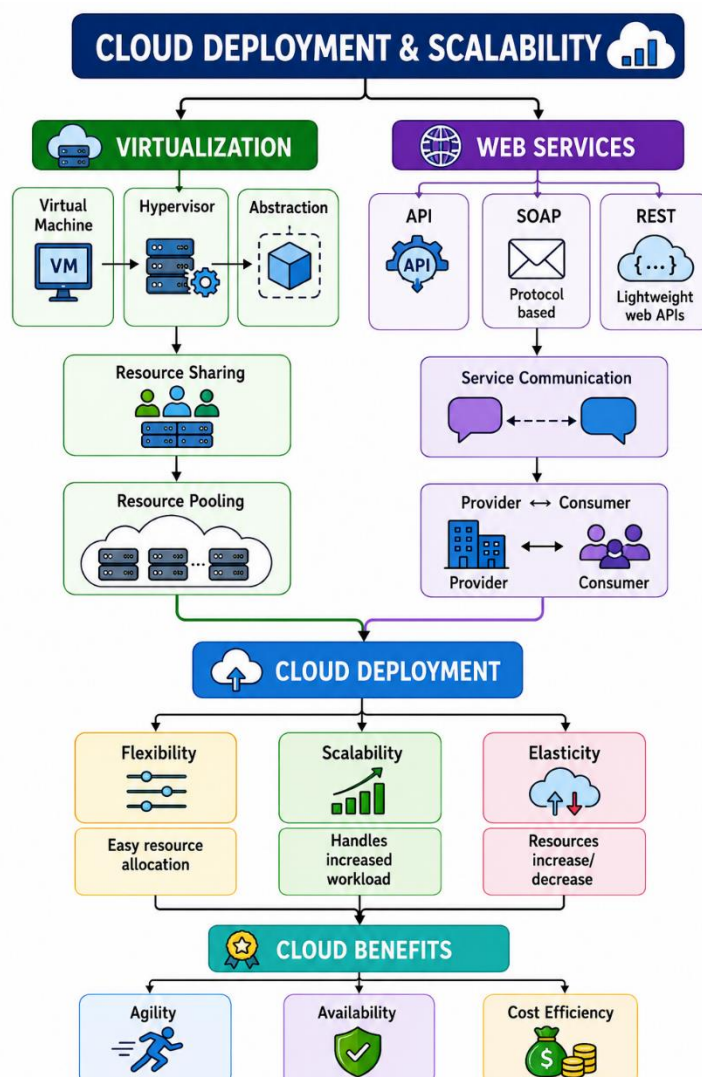
Cloud Services → Service Models → IaaS / PaaS / SaaS / XaaS → Provider + User Responsibilities

Key Linking Labels:

- IaaS **provides** → Infrastructure layer
- PaaS **provides** → Platform layer
- SaaS **provides** → Application layer
- XaaS **extends** → Cloud services
- Provider **manages** → Underlying infrastructure
- User **manages** → Data and usage
- Both **share** → Security responsibilities

5. Concept Map 4: Virtualization and Web Services as Enablers of Cloud

Main Concept: ☁ Cloud Deployment and Scalability



Virtualization Side

Physical Hardware → Hypervisor → Virtual Machines → Resource Pooling → Flexible Resource Allocation → Scalability

- **Hypervisor** manages → Virtual Machines
- **Virtual Machine** provides → Isolated computing environment
- **Virtualization** abstracts → Physical hardware
- **Abstraction** enables → Resource sharing
- **Resource pooling** supports → Scalability and elasticity

Web Services Side

Provider → API → SOAP / REST → Consumer

- **Provider** exposes → Services
- **API** enables → Communication
- **SOAP** provides → Protocol-based web service communication
- **REST** provides → Lightweight web service access
- **Consumer** accesses → Cloud services

Combined Concept

Virtualization + Web Services → Cloud Enablers → On-demand Deployment → Scalability + Elasticity